

Jacob Berger

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Education

Oregon State University — Corvallis, OR

Ph.D. & M.S. in Artificial Intelligence

Expected June 2030

Advisor: Dr. Lizhong Chen, STAR Lab. Coursework: Deep Learning, Parallel Programming, Operating Systems II, Numerical ODEs/PDEs, Math of Data Science, Analysis of Algorithms, Machine Learning.

Honors B.S. in Applied Computer Science & Computational Mathematics

Sept 2022 – June 2026

Summa Cum Laude (both degrees)

Accelerated Master's Program (graduate coursework as undergraduate). Honors degree. AI concentration, double major.

Research & Publications

Honors Thesis: Semantic Constraints in NLP

June 2025 – Present

Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)

- First-author submission under review at AAAI 2027 ("Parsimonious Decoding"). Developed novel metrics to enforce semantic constraints in probabilistic language models, improving vocabulary legality without degrading fluency.

Experience

Graduate Research Assistant

June 2026 – Present

Oregon State University, STAR Lab — Corvallis, OR

- Conducting funded research on semantic constraints in probabilistic language models under Dr. Lizhong Chen.

AI Research Engineer

March 2023 – Present

Lexel Synergetics — Portland, OR

- Co-authored and secured a Phase I SBIR grant (federal R&D funding) for NLP-based language model development.
- Architected end-to-end data pipelines for training and fine-tuning specialized large language models (LoRA, RAG).
- Deployed cloud infrastructure on AWS, GCP, and Vercel for model training and serving.

Head Mechanic

Dec 2022 – June 2026

MU Lanes and Games — Corvallis, OR

- Managed maintenance and diagnostics for electromechanical bowling lane systems.

Projects

Senior Capstone: AI for Home Automation

Sept 2025 – June 2026

- Integrated LLMs with Home Assistant for natural-language home automation. Presented at OSU Engineering Research Expo.

AuTuber — AI Stage Hand for Live Streaming

- LLM-driven agent observing webcam, screen, and audio to autonomously control OBS and VTube Studio via WebSocket. Team of 4, served as PM and technical lead.

RelCube — Deep RL Rubik's Cube Solver

- Transformer-based solver (~3.5M params) using structured piece representation. Trained with Approximate Value Iteration and Expanding Curriculum. Team of 3.

AI for Home Automation (Senior Capstone)

- Integrated LLMs with Home Assistant for natural-language home automation. Presented at OSU Engineering Research Expo.

Skills

- **Languages:** Python, C++, Java, JavaScript, Bash, x86 Assembly, MATLAB
- **Machine Learning:** PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines, Reinforcement Learning
- **Systems & Cloud:** Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), AWS (EC2/S3), GCP, Git, Vercel
- **Human Languages:** Fluent Spanish (9 years immersion)

Awards

- Putnam Math Competition: Top 3 finisher at OSU.
- Finley Academic Excellence Scholarship: \$20,000 recipient.

Extracurriculars

- **Competitive Math:** OSU Putnam and COMAP Math Modeling (implemented Cellular Automata for ecosystem resilience analysis).
- **Competitive Sports:** Founded OSU national collegiate bowling team (All-Star qualifier); Rose City Rowing Club leader (2-time national qualifier).